

Module 2

1. Fitting of a straight Line.

$$y = a + bx$$

The normal equations are

$$\sum y = na + b\sum x \quad \sim (A)$$

$$\sum xy = a\sum x + b\sum x^2 \quad \sim (B)$$

Problem 1

The following observation represent the load applied to a structural member and its deflection

Load P kN	10	20	30	40	50
Deflection δ mm	0.42	0.81	1.19	1.63	2.02

Assuming a linear relationship $\delta = a + bP$ determine the fitted equation and estimate the deflection @ a load of 60 kN.

Here Load P is independent variable X
Deflection δ is dependent variable Y.

$$n = 5$$

X Y XY x^2

0.42 4.2 100

0.81 ~~16.2~~ 400

1.19 35.7 900

1.63 65.2 1600

2.02 101 2500

6.07 222.3 5500

$$\sum x = 150 \quad \sum y = 6.07$$

$$\sum xy = 222.3 \quad \sum x^2 = 5500$$

$$(A) \Rightarrow 6.07 = 5a + 150b \quad \sim (1)$$

$$(B) \Rightarrow 222.3 = 150a + 5500b \quad \sim (2)$$

Solving (1) & (2)

$$a = \frac{1}{125}; b = \frac{201}{5000}$$

$$= 0.008 \text{ kN} \quad = 0.0402 \text{ mm}$$

$\therefore$ Best fit straight Line is.

$$\delta = 0.008 + 0.0402P$$

@ P = 60 kN

$$\delta = 2.42 \text{ mm}$$

Problem 2

The electrical resistance of a sensor was measured @ different temperatures

Temperature $T^{\circ}\text{C}$ 20 30 40 50 60

Resistance $R \Omega$ 102 107 111 116 121

Fit the relation $R = a + bT$ & estimate the resistance @ 75°C

Soln

Here independent variable $x = T$
dependent variable $y = R$.

$n = 5$			
(T) x	(R) y	xy	x^2
20	102	2040	400
30	107	3210	900
40	111	4440	1600
50	116	5800	2500
60	121	7260	3600
<u>200</u>	<u>557</u>	<u>22750</u>	<u>9000</u>

$$\sum x = 200$$

$$\sum y = 557$$

$$\sum xy = 22750$$

$$\sum x^2 = 9000$$

$$\textcircled{A} \Rightarrow \begin{aligned} 557 &= 5a + 200b \\ 22750 &= 200a + 9000b \end{aligned}$$

$$a = \frac{463}{5} = 92.6 \quad b = \frac{47}{100} = 0.47$$

$$\therefore \text{best fit line is } \boxed{R = 92.6 + 0.47T}$$

now @ $T = 75^{\circ}\text{C}$

$$\begin{aligned} \text{Resistance } R &= 92.6 + 0.47(75) \\ &= \underline{\underline{127.85 \Omega}} \end{aligned}$$

? Fitting of a second degree parabola.

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$$y = a + bx + cx^2$$

The normal eq^{ns} are.

$$\sum y = na + b\sum x + c\sum x^2 \quad \text{--- (C)}$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3 \quad \text{--- (D)}$$

$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4 \quad \text{--- (E)}$$

Problem 3 The breaking distance of a vehicle was observed @ different speeds.

Speed v kmph 20 30 40 50 60

Breaking distance d m. 5 9 15 23 33

Fit the relation $d = a + bv + cv^2$ & estimate the breaking distance @ 70 kmph.

Solⁿ

Here independent variable $x = v = \text{speed}$.

dependent variable $y = d = \text{Breaking distance}$.

Here $n = 5$

x	y	xy	x^2	x^2y	x^3	x^4	
20	5	100	400	2000	8000	160000	$\sum x = 200$
30	9	270	900	8100	27000	810000	$\sum y = 85$
40	15	600	1600	24000	64000	2560000	$\sum xy = 4100$
50	23	1150	2500	57500	125000	6250000	$\sum x^2 = 9000$
60	33	1980	3600	118800	216000	12960000	$\sum x^2y = 210400$
$\sum$	200	85	4100	9000	210400	440000	$\sum x^3 = 440000$
							$\sum x^4 = 22740000$

$$\text{C} \Rightarrow \text{C} \Rightarrow 85 = 5a + 200b + 9000c \quad \text{--- (1)}$$

$$\text{D} \Rightarrow 4100 = 200a + 9000b + 440000c \quad \text{--- (2)}$$

$$\text{E} \Rightarrow 210400 = 9000a + 440000b + 22740000c \quad \text{--- (3)}$$

$$\text{Solving (1) (2) \& (3)} \Rightarrow a = 3 ; b = -0.1 \quad \& \quad c = +0.01$$

$\therefore$ best fit second degree parabola is $y = 3 - 0.1x + 0.01x^2$

$$\text{Breaking distance } d \text{ @ } 70 \text{ kmph} = 3 - 0.1(70) + 0.01(70)^2 = \boxed{45 \text{ m}}$$

Problem 4 : The following load were obtained from a beam test.

Load P kN	0	5	10	15	20
Deflection δ , mm	0	0.28	0.67	1.19	1.85

Fit the quadratic eqⁿ $\delta = a + bP + cP^2$
 estimate deflection @ $P = 25 \text{ kN}$

Here the independent variable is Load, $x = P$
 dependent variable is deflection, $y = \delta$

$$n = 5$$

x	y	xy	x^2	x^2y	x^3	x^4
0	0	0	0	0	0	0
5	0.28	1.40	25	7	125	625
10	0.67	6.70	100	67	1000	10000
15	1.19	17.85	225	267	3375	50625
20	1.85	37	400	740	8000	160000
$\Sigma 50$	3.99	62.95	750	1081.75	12500	221250

$$\Sigma x = 50$$

$$\Sigma xy = 62.95$$

$$\Sigma x^2y = 1081.75$$

$$\Sigma y = 3.99$$

$$\Sigma x^2 = 750$$

$$\Sigma x^3 = 12500$$

$$\Sigma x^4 = 221250$$

$$\textcircled{1} \Rightarrow 3.99 = 5a + 50b + 750c \quad \sim \textcircled{1}$$

$$\textcircled{2} \Rightarrow 62.95 = 50a + 750c + 12500c \quad \sim \textcircled{2}$$

$$\textcircled{3} \Rightarrow 1081.75 = 750c + 12500c + 221250c \quad \sim \textcircled{3}$$

$$\text{Solving } \textcircled{1} \textcircled{2} \textcircled{3} \quad a = 0.003143$$

$$b = 0.041343$$

$$c = 0.002543$$

$\therefore$ Best fit second degree parabola is.

$$\delta = 0.003143 + 0.041343P + 0.002543P^2$$

For $P = 25 \text{ kN}$, deflection $\delta =$

$$\delta = 0.003143 + 0.041343(25) + 0.002543(25)^2 = 2.626 \text{ mm}$$